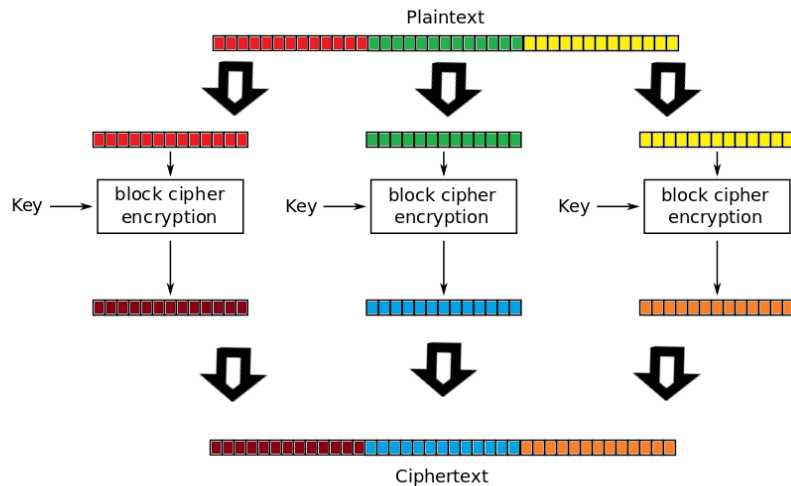


Crypto Training Challenges Day 2

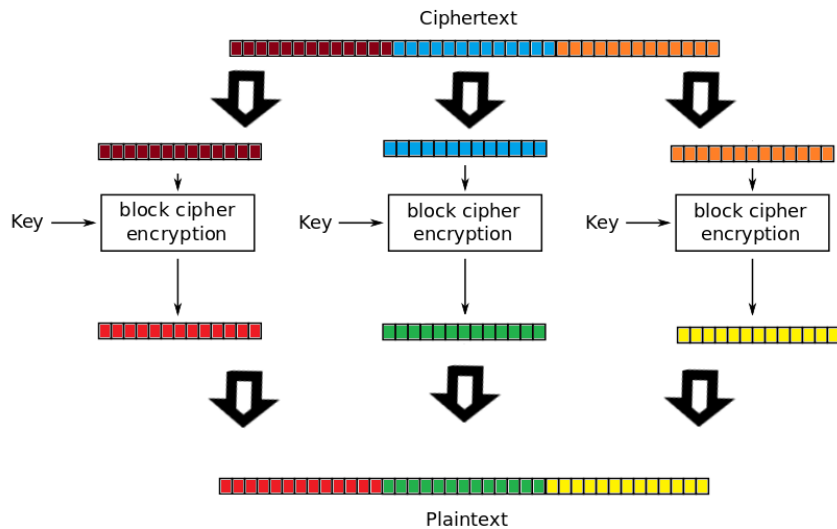
All these challenges are designed for <https://gchq.github.io/CyberChef>

1. AES ECB (Electronic Code Book)

Electronic Codebook (ECB) mode encryption



Electronic Codebook (ECB) mode decryption



Here are 7 128-bit blocks of data encrypted with AES-ECB in Hex:

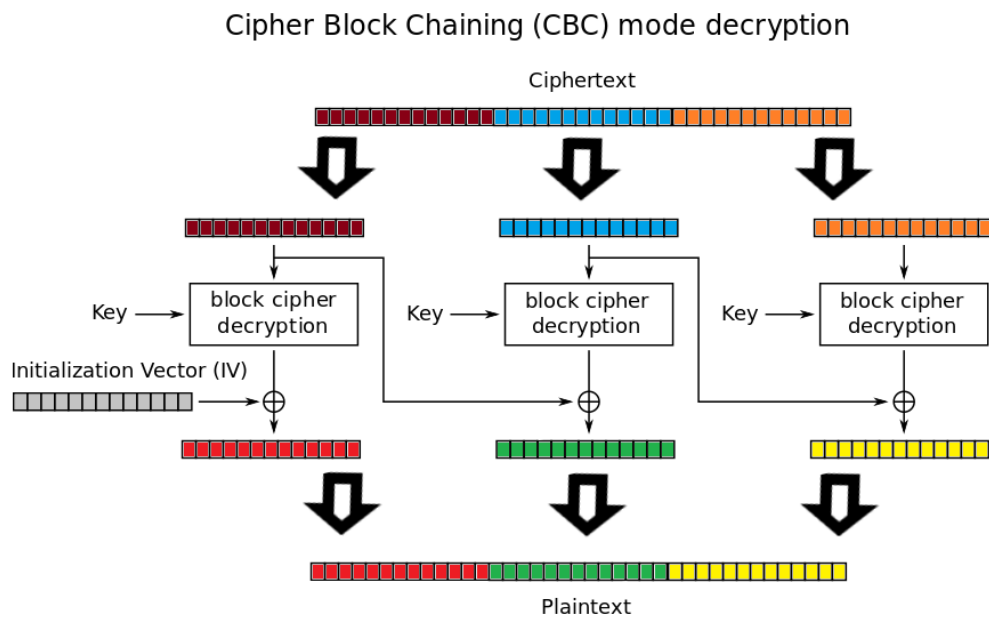
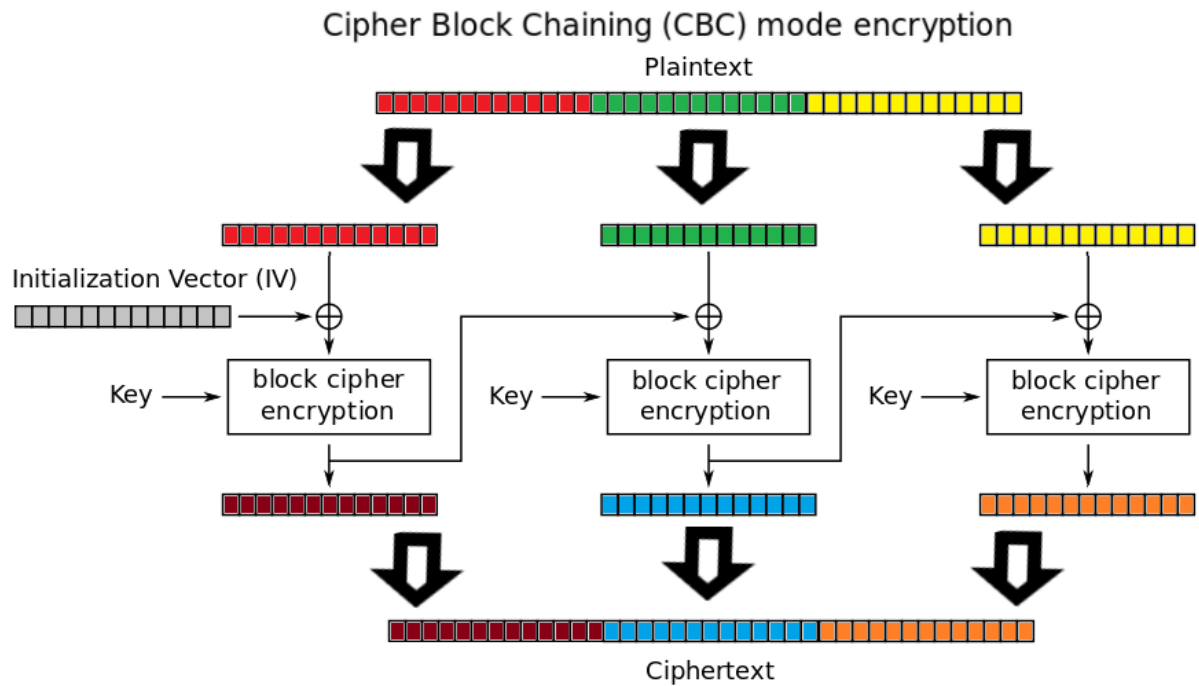
```
0b54f4ac504af60a228d4038faa3d797
d5d3a6ca3e451b091a4e9d2648c81dc4
21f5c270d1c70602ffd454a87e63424e
3bb61cd8d6110d9804365553a3e4dc3c
b3bed86a1a33b78ff26ab9b39d4f6a78
251ba614795db8f5bfb049bf061a9690
d78fb08167f08e88b17583e70b96323d
```

In this case we know the used key: PASSWORDPASSWORD
(this password is the worst :))

!!Just modifying the whole Plaintext and encrypting it would work too but it is not the point here.!!!

1. Use Cyberchef AES decrypt to gain the plain text. It may contain something interesting.
2. Duplicate one of the blocks in the ciphertext. Does it work?
What is the result?
3. Delete another block. Anything out of the ordinary?
4. Switch out the Block that contains the credit card number with a different arbitrary number and switch out the amount too.
Do this by only encrypting your new data and switch out the single blocks.

2. AES CBC (Cipher Block Chaining)



Again 7 blocks of 128 bits:

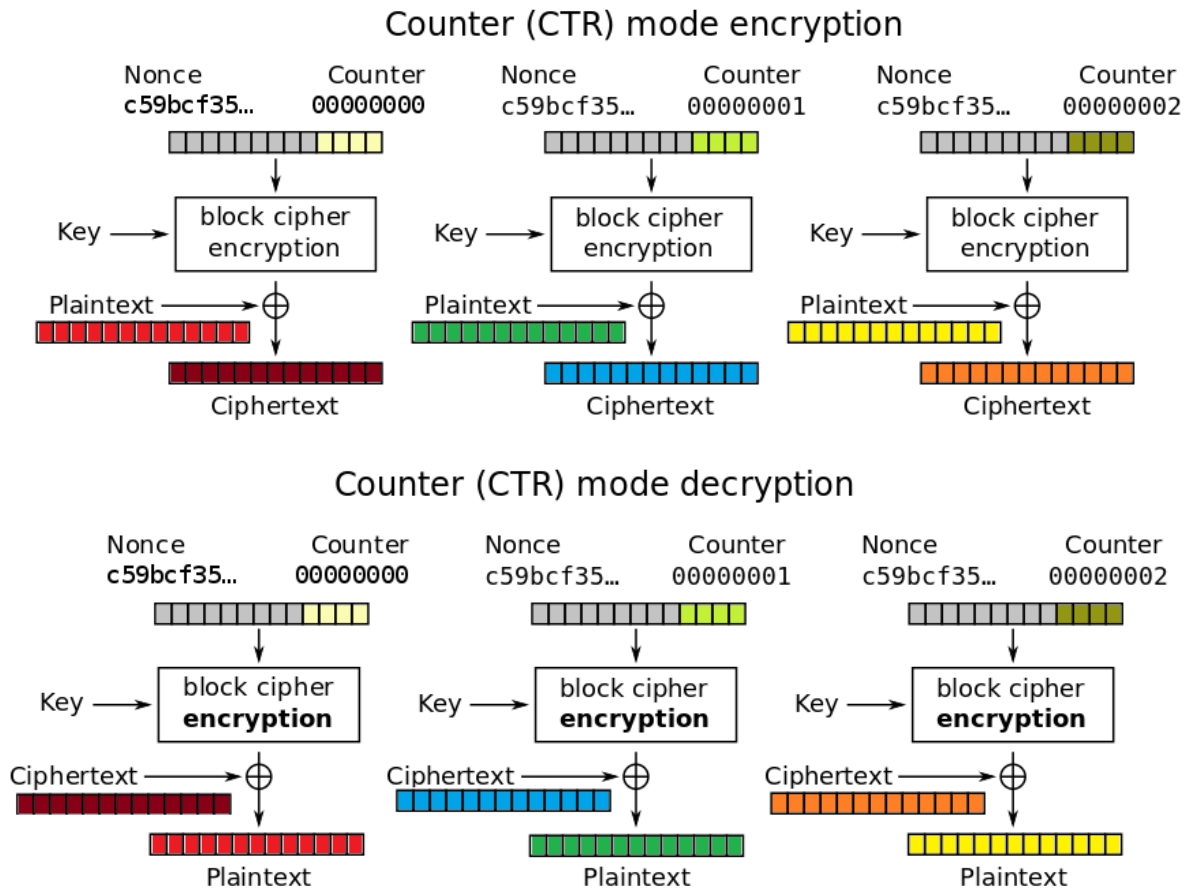
```
cc7ac51e8b43d0a86368669e3d01be44
450beff03243567f490b1a69d40e48b5
1d946a862d65d9b60ff560487ec7022f
7c6230a03ae657203edfb20deabeaf84
4e2582fd64065196142a2c52f2fabf21
363cb328ba216560d9e447f1175282b2
724002bef489f10b5c2fd472c61b9461
```

Password: PASSWORDPASSWORD and IV: ABCDEFGHIJKLMNOP

!!Do this by only encrypting your new data and switch out the single blocks. Just altering the whole Plaintext and encrypting it would work too but it is not the point here.

1. Use Cyberchef AES decrypt to gain the plain text. It is the same as last time but now with cbc.
2. Duplicate one of the blocks in the ciphertext. Does it work? How many blocks are impacted?
3. Delete another block. Anything out of the ordinary? How many blocks are impacted?
4. Try to sneak in an altered Credit card number like with ECB.
5. What happens if you switch out single characters in the IV?

3. AES CTR (Counter Mode)



Again 7 blocks of 128 bits:

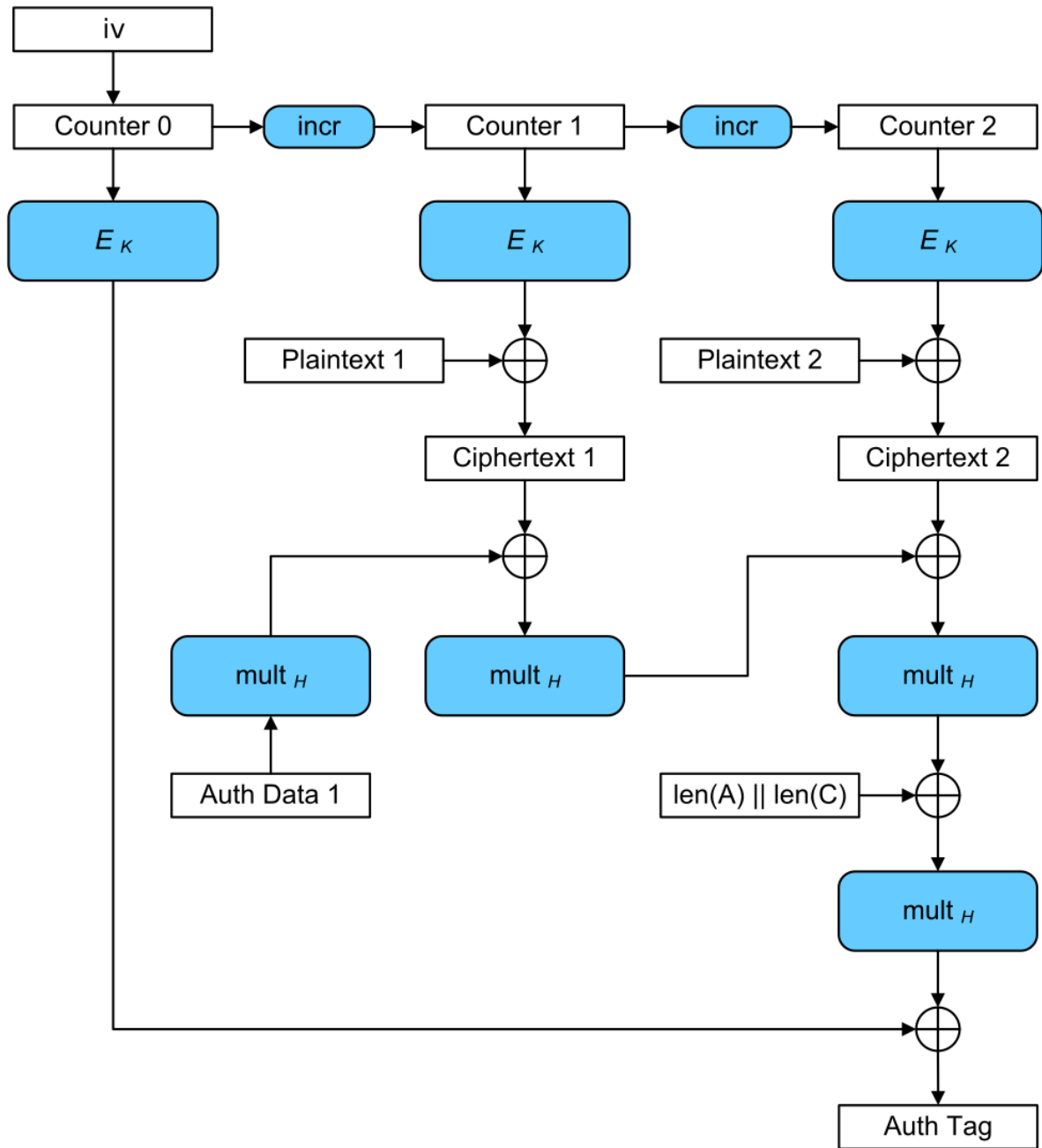
61e4173a0b3f5118938207457cc7915d
b8813cc5ed787c51d41aeb7f0f2dd299
b913aeff0392836e5a9a3ffc9ac37ca1
c9f008cb7b5f6b40e7a03e502d52d6d4
0d07e2e799488e3a57e098ab6ef1ca77
c08e836f415bf7b8f2c3002aa2c93237
f5ab179572bd12b5ceb960ce4d3442e7

Password: PASSWORDPASSWORD and IV: ABCDEFGHIJKLMNOP

!!Do this by only encrypting your new data and switch out the single blocks. Just altering the whole Plaintext and encrypting it would work too but it's not the point here.

1. Use Cyberchef AES decrypt to gain the plain text. It's the same as last time but now with CTR.
2. Duplicate one of the blocks in the ciphertext. Does it work? How many blocks are impacted?
3. Delete another block. Anything out of the ordinary? How many blocks are impacted?
4. Try to sneak in an altered Credit card number like with ECB. Did it work?

4. AES GCM (Galois Counter Mode)



Again 7 blocks of 128 bits:

70ff9b5bce4b7788ac5d196a5416cdfc
81ce101a6dbf6d071a551edfd5486b9c
7ea554dd2ef0db1dc588a3af4f1a5398
3bb61cd8d6110d9804365553a3e4dc3c
b382870b1ac2a7dc8112f7c738e1ceef
e5c1d5451e46f638fc2176d351ea507c
76a43d393c97f0b405fa2d14d3e6a3bb
cf931b633902306b319e90dc0f9e1684

This time we got an additional Tag to check if everything is all right
Tag: b274cf8d0ba0cc941a7a7975680fa75d

Password: PASSWORDPASSWORD and IV: ABCDEFGHIJKLMNOP

!!Do this by only encrypting your new data and switch out the single blocks. Just altering the whole Plaintext and encrypting it would work too but it's not the point here.

1. We have now a tag to check for integrity. Use Cyberchef to verify the integrity of the ciphertext with the tag. Does it decipher with no error?
2. Oh no the cyphertext was probably modified!? May be someone just tried to sneak a block in. Try to find the malicious block and remove it.
3. Recheck the tag to make sure that everything is fine. Also, it should now decrypt correctly.
4. Try to delete another block and see what happens.